

Experiment 3: Summary Table

Dropped-Only Quality

Method	N(drop)	Precision Mean(m)	Precision P50(m)	Precision P90(m)	R@1m	R@2m	R@3m	Roll Mean	Roll P90	Pitch Mean	Pitch P90	Impact Mean	Impact P90	Impact P95
PPO	1993	11.243	13.602	21.726	15.4%	30.4%	35.3%	8.170	19.271	5.000	10.451	0.417	0.980	1.273
Proposed	1962	2.314	0.946	4.264	52.8%	80.8%	87.2%	15.617	29.117	7.646	17.495	0.541	1.271	1.453
PPO-GRU	1982	2.692	1.102	8.642	45.5%	74.0%	84.3%	14.623	29.085	7.536	15.343	0.469	1.217	1.421
MPC	1049	5.634	2.869	15.707	2.8%	18.1%	56.0%	2.597	3.663	1.529	2.734	0.145	0.147	0.174

No-Drop Breakdown (done but no release)

Method	N(done)	N(drop)	Drop Rate	N(no_drop)	No-drop Rate	Crash Rate	Crash+NoDrop	Crash in NoDrop	Timeout+NoDrop	Both Flags
PPO	2000	1993	99.7%	7	0.3%	0.3%	7	100.0%	0	0
Proposed	2000	1962	98.1%	38	1.9%	1.9%	38	100.0%	0	0
PPO-GRU	2000	1982	99.1%	18	0.9%	0.9%	18	100.0%	0	0
MPC	2000	1049	52.5%	951	47.5%	47.5%	951	100.0%	0	0

Cross-Case Check: PPO/Proposed/PPO-GRU on MPC No-Drop Cases (large error > 3.0m)

Method	In MPC-NoDrop	Dropped	Drop Rate	Err>3.0m	LargeErr Rate (in dropped)	Mean Err (in dropped)
PPO	951	946	99.5%	664	70.2%	12.058
Proposed	951	929	97.7%	145	15.6%	2.451
PPO-GRU	951	937	98.5%	138	14.7%	2.700